Ontology Engineering

Session 5

Assumptions

- Febraury 18
  - **Protégé** installed, and basic knowledge of its functionalities
- February 25
  - **formal taxonomy** of road signs as an IS-A class hierarchy, written in OWL2
  - conformant to Part I, Annex 1, of Vienna Convention (document “Conv_road_signs_2006v_EN.pdf”)
- March 4
  - **extended taxonomy**, aligned with the category labels of the RDF knowledge graph of the Mapillary Traffic Sign Dataset (file “MTSD.ttl”)
- March 18
  - **object properties and data properties**
  - conformant to Part I, Annex 1, of Vienna Convention (document “Conv_road_signs_2006v_EN.pdf”)
- March 25
  - **property characteristics** (for object and data properties)
  - **definitions** for *all* road-sign classes of the main hierarchy

Competency Questions

- **CQs about the taxonomy of road signs**
  - What are the primary kinds of road signs distinguished in the Vienna Convention?
  - Is a priority sign considered a type of road sign?
- **CQs about road sign properties**
  - Which border colours can circle-shaped road signs have?
  - What are the required background and border colours for a model B2a "STOP" sign?
  - What is the standard measurement for the side of a normal-sized danger warning sign?
- **CQs about the pragmatics of road signs**
  - Which danger warning symbol (A11a or A11b) must be used to warn of falling rocks?
  - What specific symbol is used on sign A15 to warn of "Cattle or other animals crossing”?
  - Which road signs are relevant to pedestrians?
- **CQs relevant to the particular dataset**
  - How many road signs are depicted in the image IMG_20230314_49094.jpg?
  - Which images contain a “COMPULSORY MINIMUM SPEED” sign?
  - What percentage of road signs depicted in the image repository are prohibitory signs?
  - Is there any road sign in the image repository that informs about a restriction applicable to lorries or other large vehicles?
  - What is the maximum speed shown on most speed limit signs?

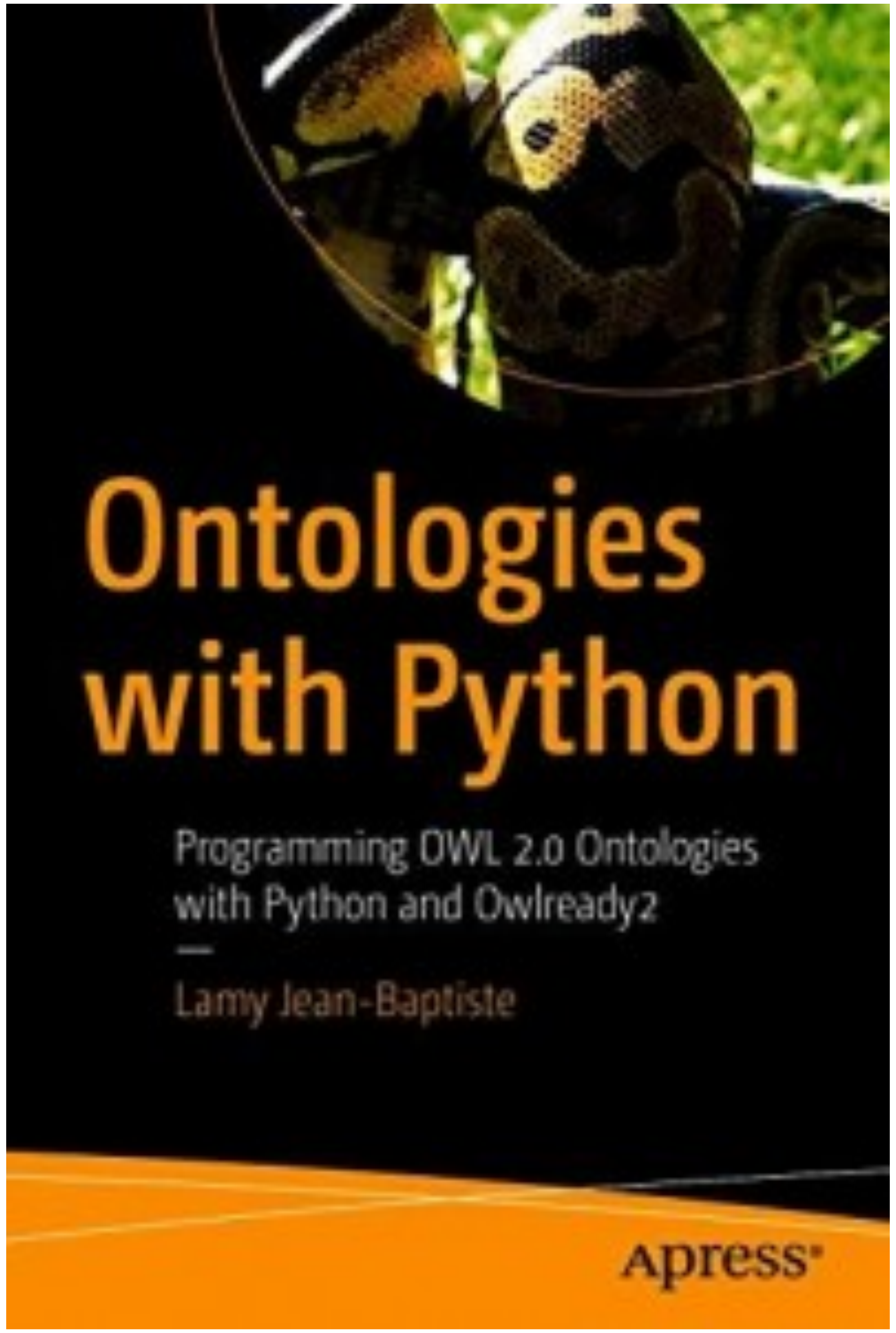
Assignment

- Check your ontology with the Semantic Reasoner (Hermit or Pellet)
  - **consistency** and **inferred axioms**
  - **competency questions**:
    - by way of **DL queries**
    - by way of **SPARQL queries**
    - by way of **Python scripts** (using Owlready2)
- Generate queries/scripts with the appropriate prompts for LLM-based formalisation

LLM-generated Python script for: “What shapes are associated with each traffic sign category?”

```
from owlready2 import *

# ------------------------------------------------------------
# Load ontology
# ------------------------------------------------------------
onto_path.append(".")  # current directory
onto = get_ontology("roadsigns3.rdf.xml").load()

# ------------------------------------------------------------
# Run a reasoner
# ------------------------------------------------------------
# Pellet is bundled with Owlready2 and supports OWL 2 DL reasoning
with onto:
    sync_reasoner_pellet(infer_property_values=True, infer_data_property_values=True)

# ------------------------------------------------------------
# Helper: get individuals or classes with a given restriction
# ------------------------------------------------------------
def get_shapes_for_category(category):
    """
    Returns the set of shapes associated with a traffic sign category.
    A shape is associated if it is entailed via:
      - hasShape value X
      - hasShape some X
    """
    shapes = set()

    # Iterate over all subclasses (including the category itself)
    for cls in category.descendants(include_self=True):
        for restriction in cls.is_a:
            if isinstance(restriction, Restriction) and restriction.property == onto.hasShape:
                value = restriction.value
                if value is not None:
                    shapes.add(value)

    return shapes

# ------------------------------------------------------------
# Traffic sign categories (rooted at RoadSign)
# ------------------------------------------------------------
categories = [
    onto.DangerWarningSign,
    onto.PrioritySign,
    onto.ProhibitoryOrRestrictiveSign,
    onto.MandatorySign,
    onto.SpecialRegulationSign,
    onto.InformationFacilityServiceSign,
    onto.DirectionPositionIndicationSign,
    onto.AdditionalPanel,
]

# ------------------------------------------------------------
# Compute and print results
# ------------------------------------------------------------
print("Shapes associated with each traffic sign category:\n")

for category in categories:
    shapes = get_shapes_for_category(category)

    print(f"- {category.name}:")
    if shapes:
        for shape in sorted(shapes, key=lambda s: s.name):
            print(f"    • {shape.name}")
    else:
        print("    • (no shape entailed)")
    print()
•
```

Owlready2


Ontologies with Python
Programming OWL 2.0 Ontologies with Python and Owlready2
Lamy Jean-Baptiste
Apress

For this step

- Competency questions and their formalisation as DL queries, SPARQL queries, or Python scripts
- LLMs and prompts used
- Answers to the competency questions

Optional Submission

- List of **competency questions** that is guiding your ontology developments

- Current state of your **road-sign ontology** with the main taxonomy of road signs, MTDS categories, object and data properties (with their characteristics) and road-sign definitions